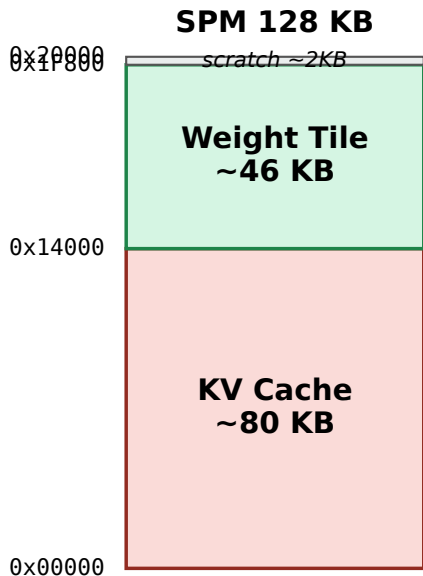


(a) SPM partition



(b) Sizing analysis

KV cache sizing (Multi30K Transformer)

Parameters:

$d_{model} = 128$ (4 heads $\times$ 32)

$L = 2$ decoder layers

$T_{max} = 50$ tokens (Multi30K)

bytes = 2 (FP16)

Per-layer K size = $T_{max} \times d_{model} \times 2 = 12.8$ KB

Per-layer V size = $T_{max} \times d_{model} \times 2 = 12.8$ KB

Total KV = $2(K+V) \times L = 2 \times 25.6 \times 2 = 51.2$ KB

Reserved budget (incl. encoder ctx & double buffer):
→ 80 KB (1.56 $\times$ headroom)

Weight tile buffer

Largest GEMV tile: $512 \times 64 \times 2$ B = 64 KB

Streaming double-buffer: $32 + 32 = 46$ KB tile

Fig. 4. SPM 128 KB partition with KV-cache sizing.